



Windows 11 installation

By Albert Zhang

Installation of Windows 11

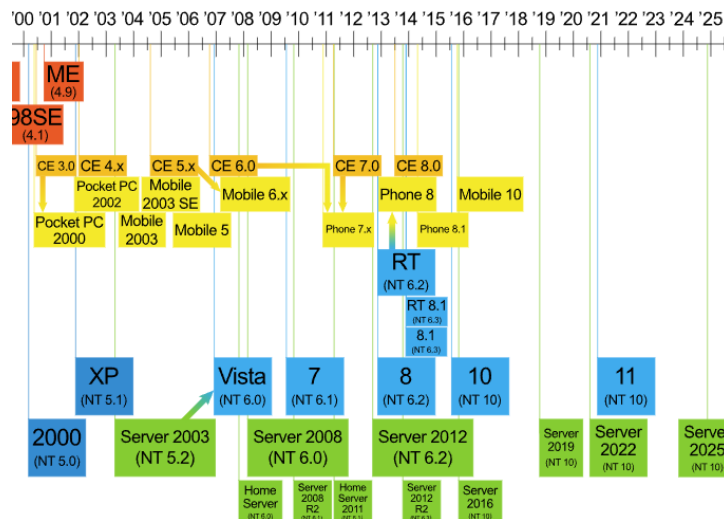
Purpose:

Installation and initial configuration of **Windows 11** on a new or reformatted device. This includes not only the operating system installation but also the setup of essential applications commonly used in educational, professional, and networking environments.

Background:

Windows is a product line of proprietary graphical operating systems developed and marketed by Microsoft. The name "Windows" is a reference to the windowing system in GUIs (graphical user interfaces). The first version, Windows 1.0 was released on November 20 1985, it was developed as a graphical shell for MS-DOS in response to the increasing demand for GUIs.

Development of windows versions after the 2000s:



The most recent version of Windows is Windows 11 for consumer PCs and tablets, Windows 11 Enterprise for corporations. Windows 11 comes with a wide range of new feature



Windows 11, released on October 5, 2021, is Microsoft's latest major

Windows NT operating system, succeeding Windows 10 (2015). It's a free upgrade for compatible Windows 10 devices. Its server counterpart, Server 2025, launched in 2024, Windows 11 comes with numerous key features such as:

Widgets Panel: A customizable feed of information like news, weather, and other personalized content is easily accessible from the taskbar.

Microsoft Teams Integration: Teams chat is integrated directly into the taskbar for easier communication with contacts.

Redesigned Microsoft Store: The Store offers better experience for both users and developers, with a wider range of apps and more options for developers to publish their apps.

Updated Default Apps: Many built-in apps have been updated, including the Photos app for better photo management and the Paint app for photo and video editing.

Redesigned Settings App: A new Settings app provides a more streamlined and user-friendly way to manage your PC settings.

Improved Multi-Monitor Support: Windows 11 offers improved handling of multiple monitors, making it significantly easier and more efficient to work with several displays simultaneously.

Enhanced Security Features: Windows 11 has improved security measures, including a requirement for a Trusted Platform Module (TPM) 2.0 for better hardware security.

Essential Applications and drives installed for optimal performance during this lab:

Packet Tracer:

A free network simulation software by Cisco Systems that allows users to design, build, and test virtual network topologies in a risk-free environment. It supports a wide range of Cisco and other network devices, enabling users to practice configuring equipment, learning network protocols, and troubleshooting to prepare for Cisco certifications, especially CCNA.

NVIDIA Drivers:

Nvidia Corporation is an American technology company headquartered in Santa Clara, California. Founded in 1993 by Jensen Huang (president and CEO), Chris Malachowsky, and Curtis Priem, it develops graphics processing units (GPUs), systems on chips (SoCs), NVIDIA drivers are essential software that allow your operating system to communicate with your NVIDIA GPU, ensuring optimal performance, stability, and security. They provide crucial performance improvements

Wireshark:

Wireshark is a free, open-source network protocol analyzer used to capture and analyze network traffic in real-time, allowing the user to inspect network packets for troubleshooting, security analysis, protocol debugging, and network performance optimization.

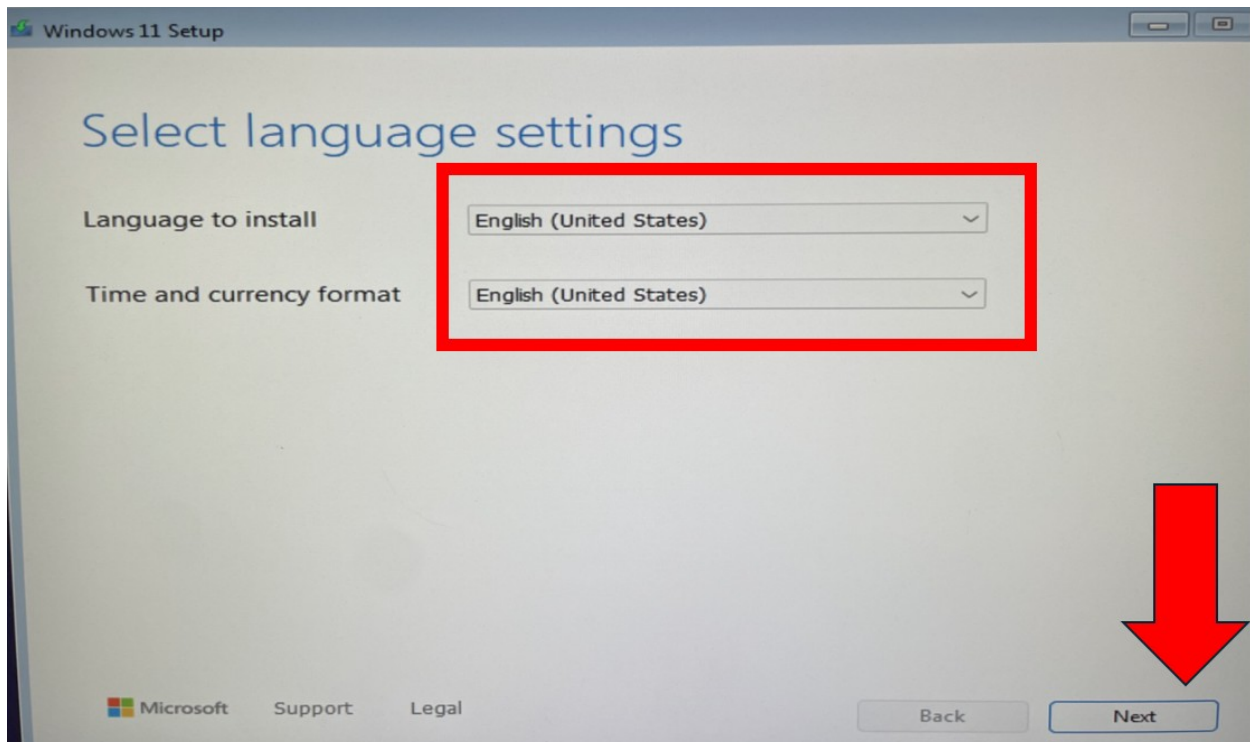
Microsoft 365:

Microsoft 365 (formerly Office 365) is a subscription-based service that offers a suite of cloud-powered productivity tools, including Office apps such as: Word, Excel, and PowerPoint, along with cloud services and other applications like OneDrive and Microsoft Teams, essential for school and work environments.

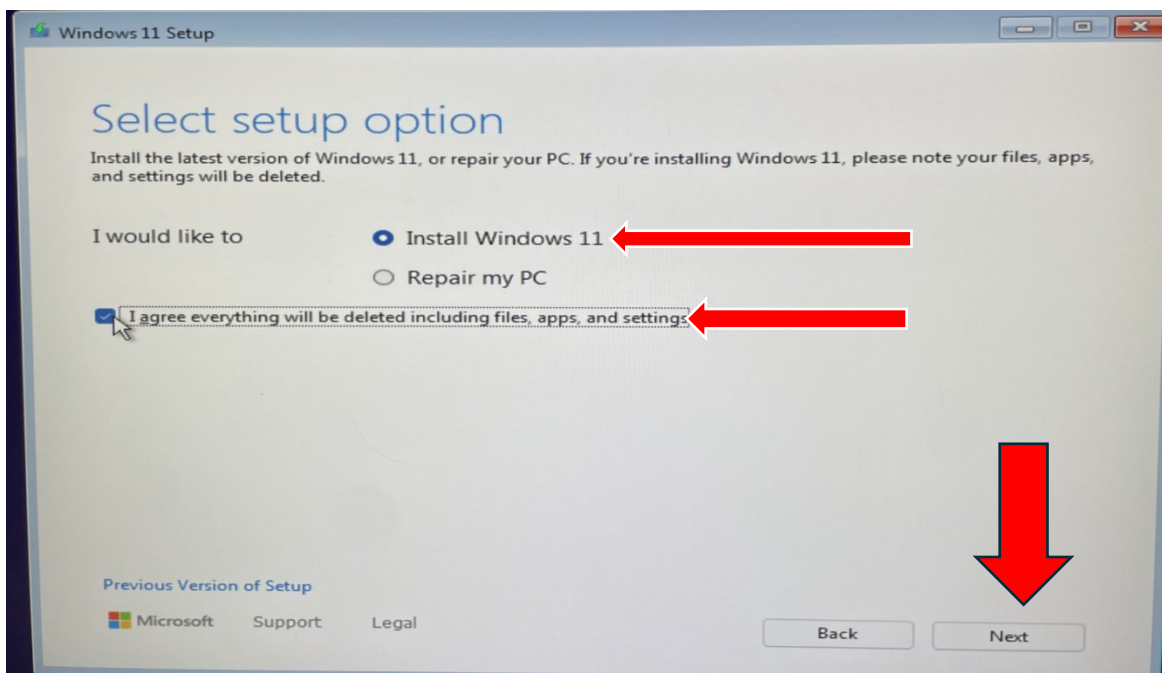
Configuration:

Installing Windows:

1. First, insert drive with Windows 11 on it, and you will be prompted to select a language. Select Language and click "Next"



2. After Selecting the language and clicking next, you will be prompted with a setup option, in the case of installing windows, select "Install Windows 11" and select agree to the prompt in the bottom. Click "Next"



3. After the setup option is selected, you will need to accept the license terms. Click on "Accept"

Applicable notices and license terms

Please read this so you know what you're agreeing to.

Your use of this software is subject to the terms and conditions of the license agreement by which you acquired this software. If you are a volume license customer, use of this software is subject to your volume license agreement. You may not use this software if you have not validly acquired a license for the software from Microsoft or its licensed distributors.

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Decline

Accept

4. Next you will be able to select the location to install Windows 11, preferably on unallocated space. Click "Next"

Select location to install Windows 11

Refresh

Load Driver

Bring Disk Online

Delete Partition

Format Partition

Create Partition

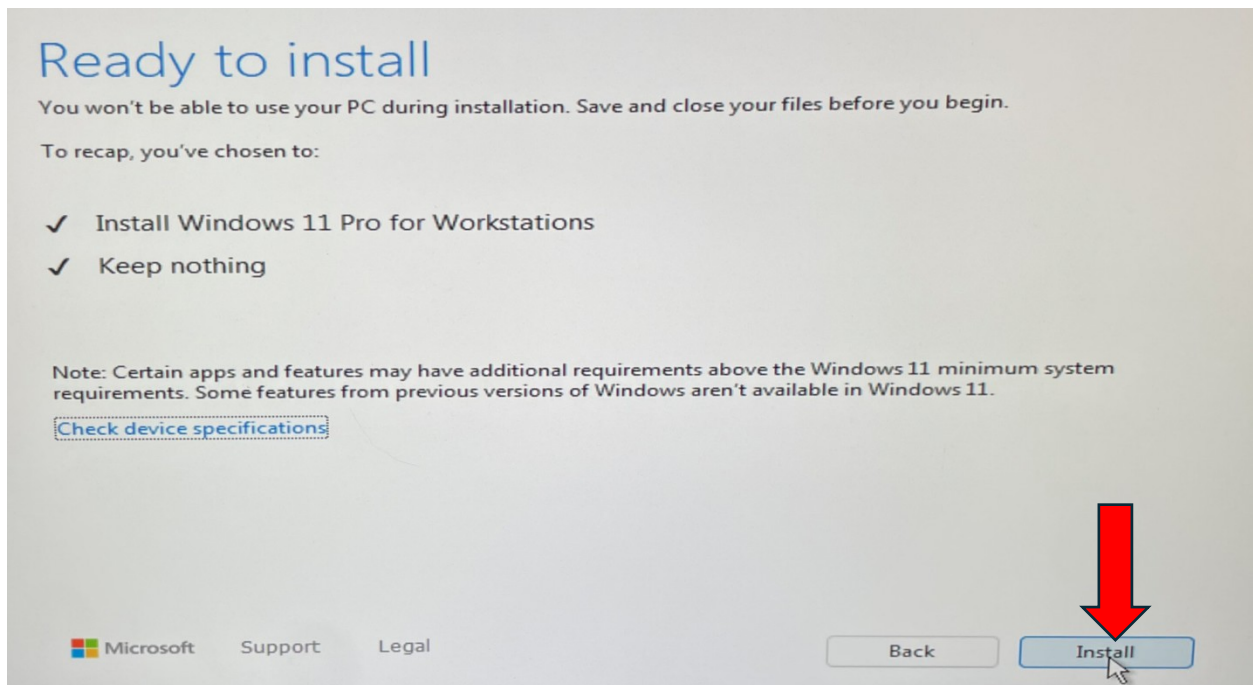
Extend Partition

Name	Total Size	Free Space	Type
Disk 0 Unallocated Space	953.9 GB	953.9 GB	Unallocated Space
Disk 1 Partition 1: CCCOMA_X64FRE_EN-US_DV9	59.8 GB	54.2 GB	Primary
Disk 1 Unallocated Space	19.0 KB	19.0 KB	Unallocated Space
Disk 1 Partition 2: UEFI_NTFS	1.0 MB	130.0 KB	Primary
Disk 1 Unallocated Space	23.0 KB	23.0 KB	Unallocated Space

Microsoft Support Legal

Next

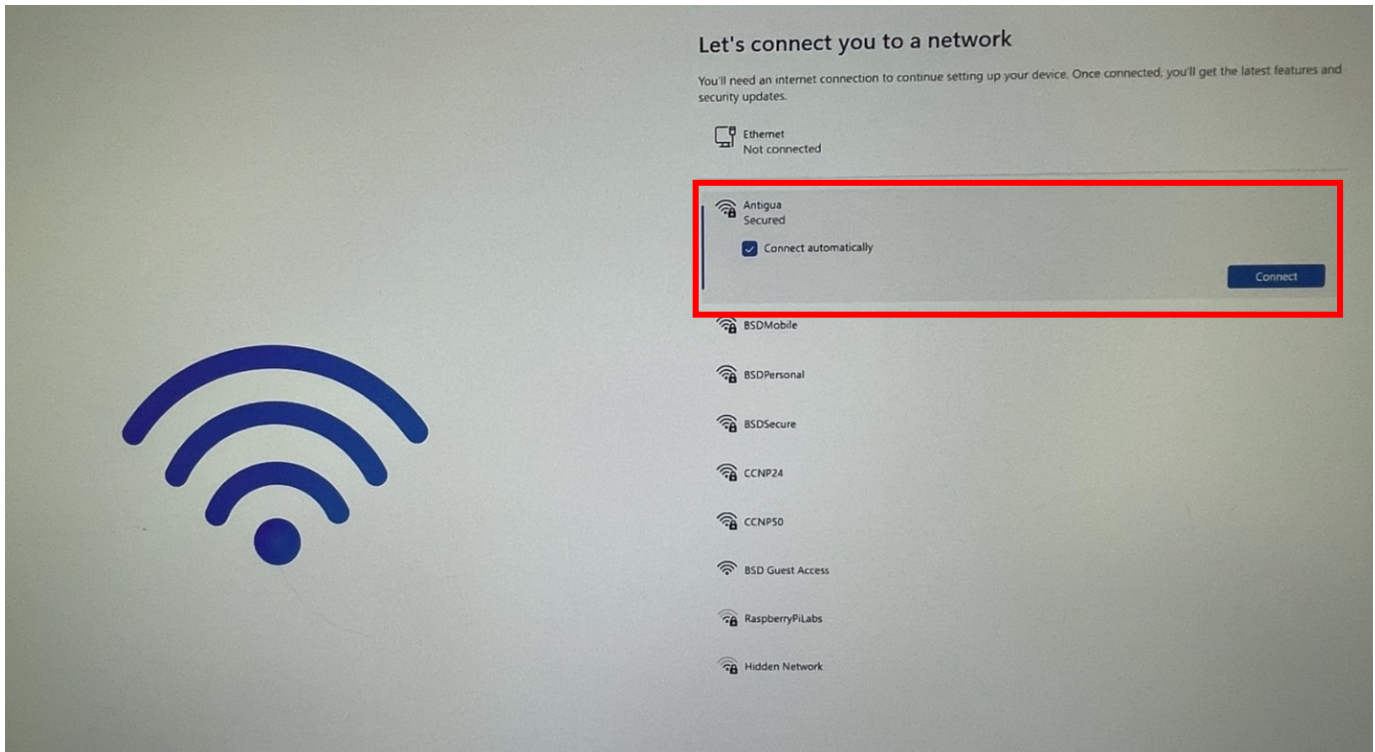
5. Now you are ready to click install. Click "Install"



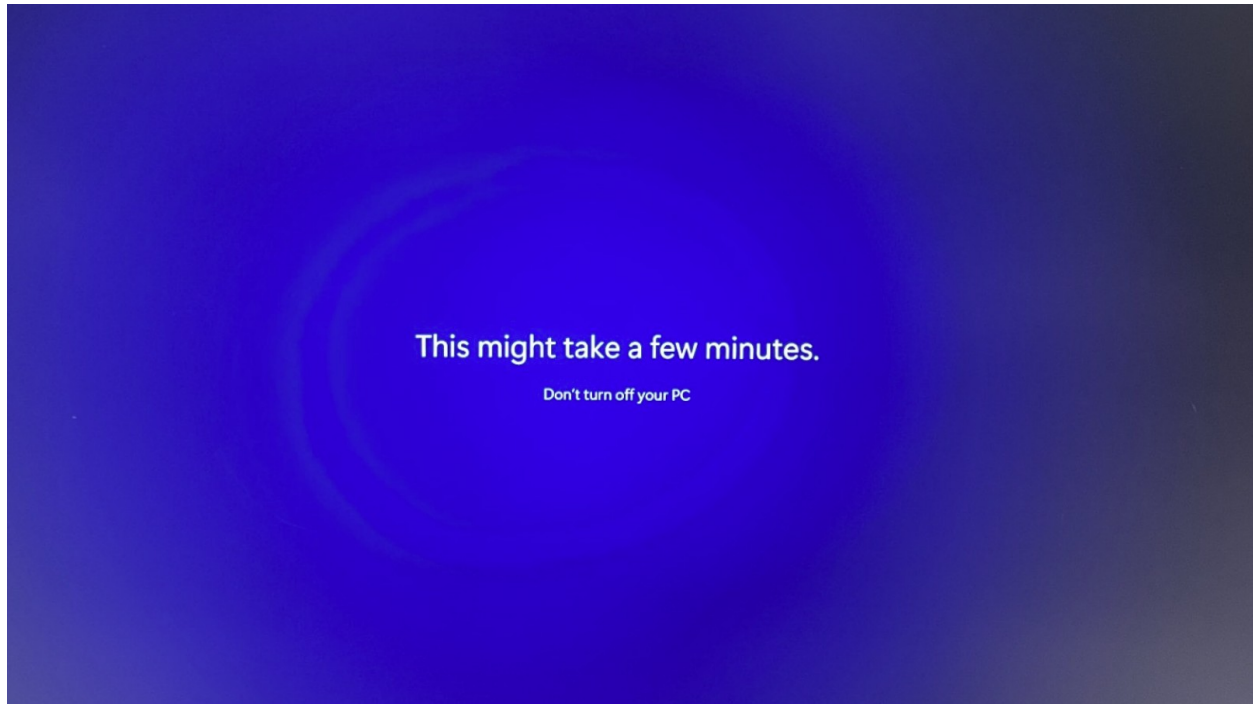
Screen you will see after clicking install (Do not click on cancel):



6. After the installation is finished, you will be prompted to connect to a network



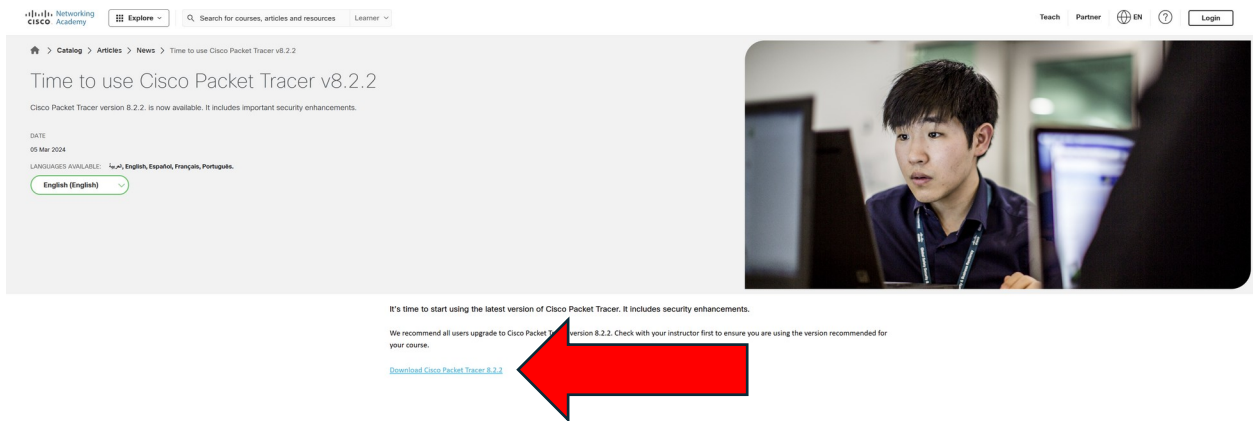
7. After connecting to a network, you will see this screen, and Windows will be installed



Installing Packet Tracer:

Go to link -> <https://www.netacad.com/articles/news/download-cisco-packet-tracer>

Then click on the download



Then you will be directed to the download page, select the download for your operating system. Note: you will be required to login to download Packet Tracer



Learning Resources



Cisco Packet Tracer

Cisco Packet Tracer, an innovative network configuration simulation tool, helps you hone your networking configuration skills from your desktop. Use Packet Tracer to experiment while building, managing & securing infrastructures.

To obtain and install your copy of Cisco Packet Tracer, please follow these simple steps:

Step 1: Download the version of Packet Tracer you require.

[Packet Tracer 8.2.2 MacOS 64bit](#)

[Packet Tracer 8.2.2 Ubuntu 64bit](#)

[Packet Tracer 8.2.2 Windows 64bit](#)

Step 2: Launch the Packet Tracer install program.

Step 3: Launch Cisco Packet Tracer by selecting the appropriate icon.

Step 4: When prompted, click on Skills For All green button to authenticate.

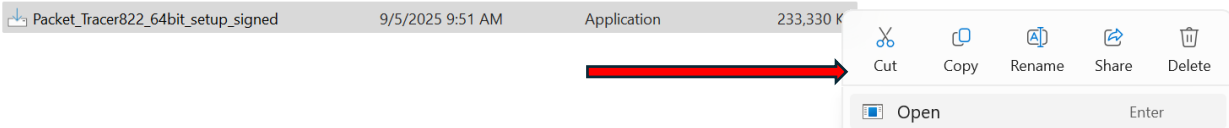
Step 5: Cisco Packet Tracer will launch and you are ready to explore its features.

If you need more guidance, please follow the [Cisco Packet Tracer Download and Installation Instructions](#).

System Requirements:

Computer with either Windows (10, 11), MacOS (10.14 or newer) or Ubuntu (20.04, 22.04) 11-bit operating system, amd64(x86-64) CPU, 4 GB of free RAM, 1.4 GB of free disk space

The file that will be downloaded:



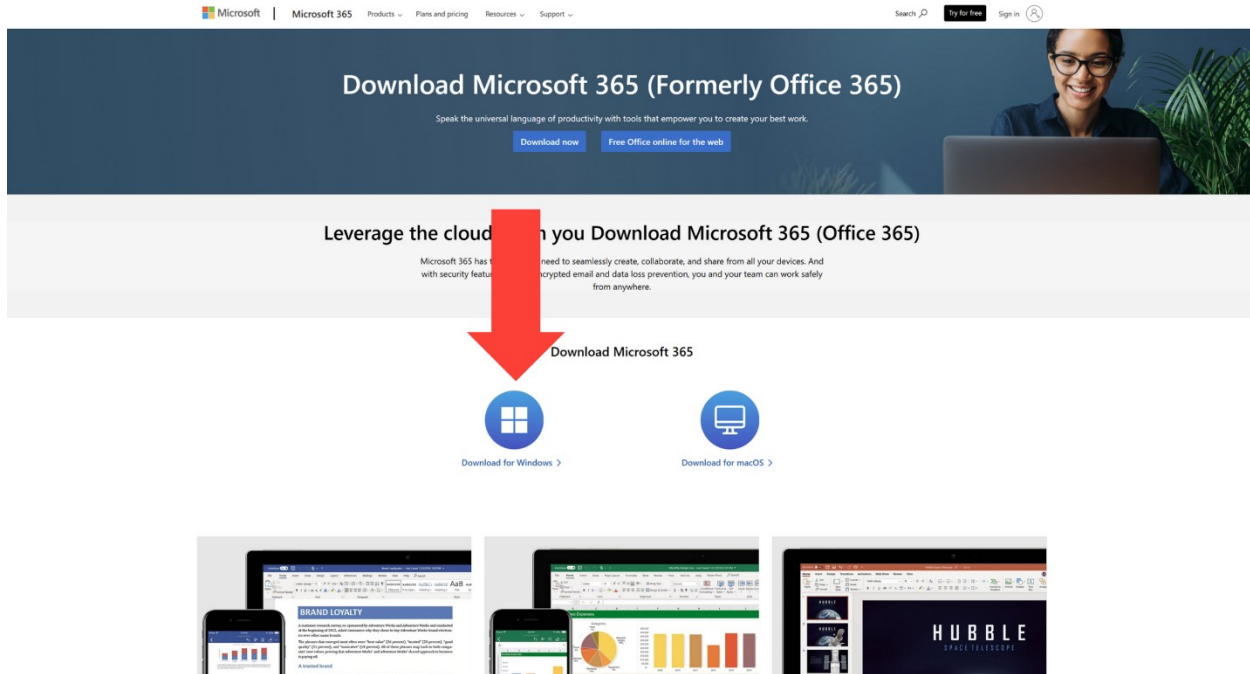
Installing PuTTY:

Search “PuTTY” in the Microsoft store, click “Install” when you find PuTTY by Simon Taham

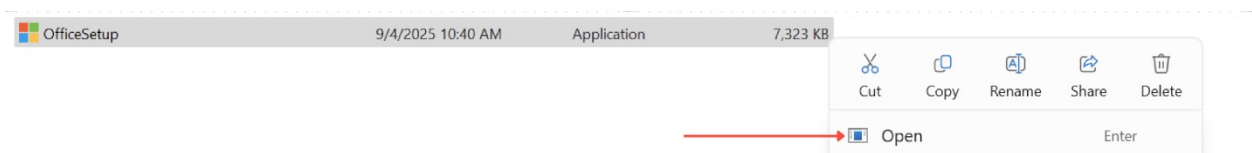


Installing Microsoft 365 (Office 365):

Go to link <https://www.microsoft.com/en-us/microsoft-365/download-office>

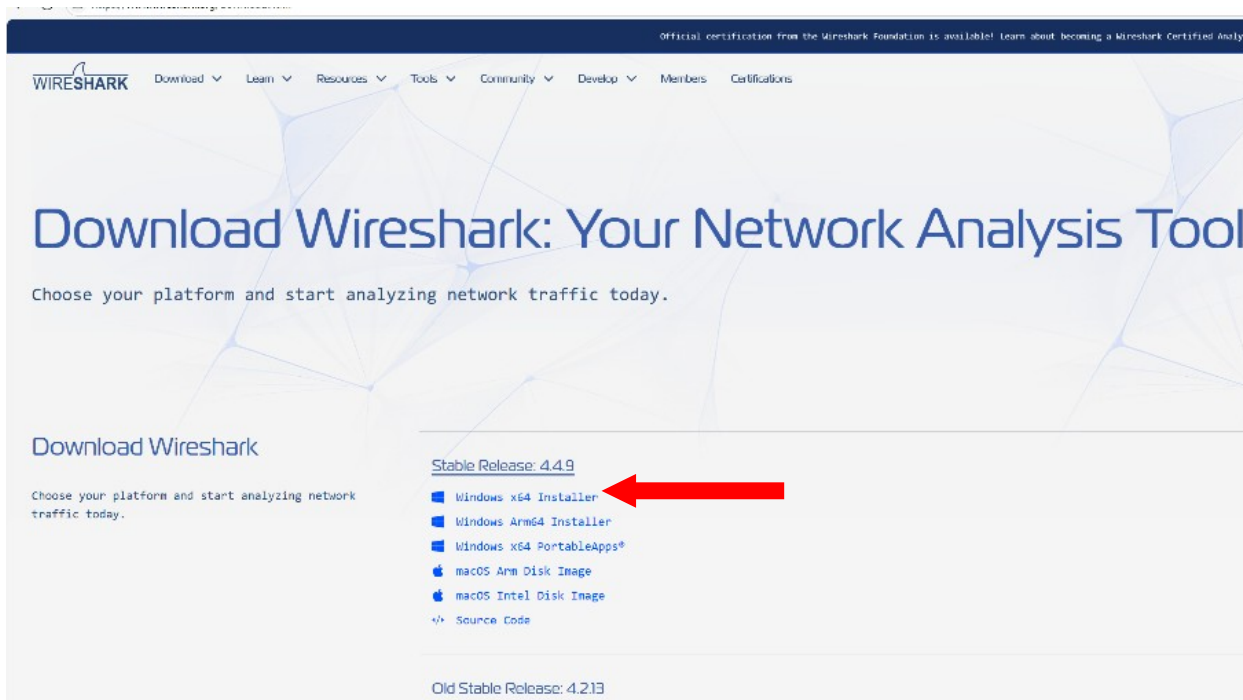


Open the downloaded file (Office Setup)

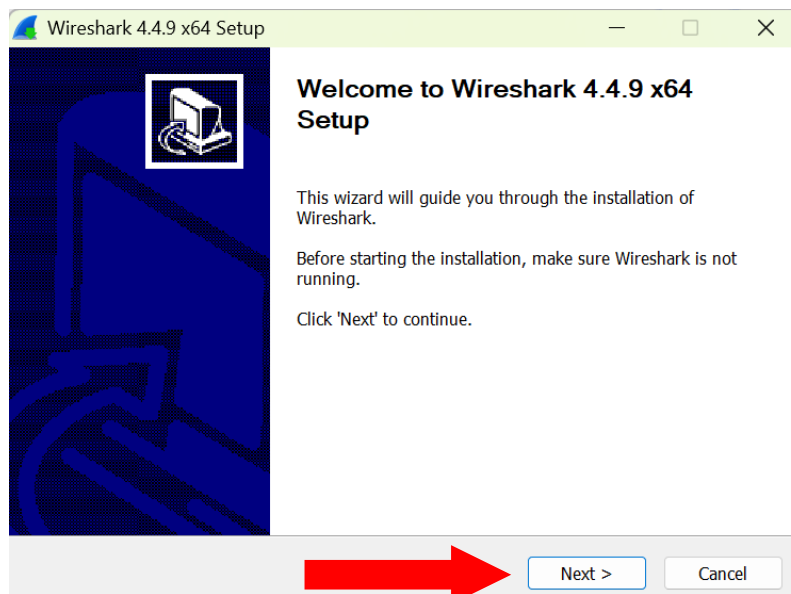


Downloading Wireshark:

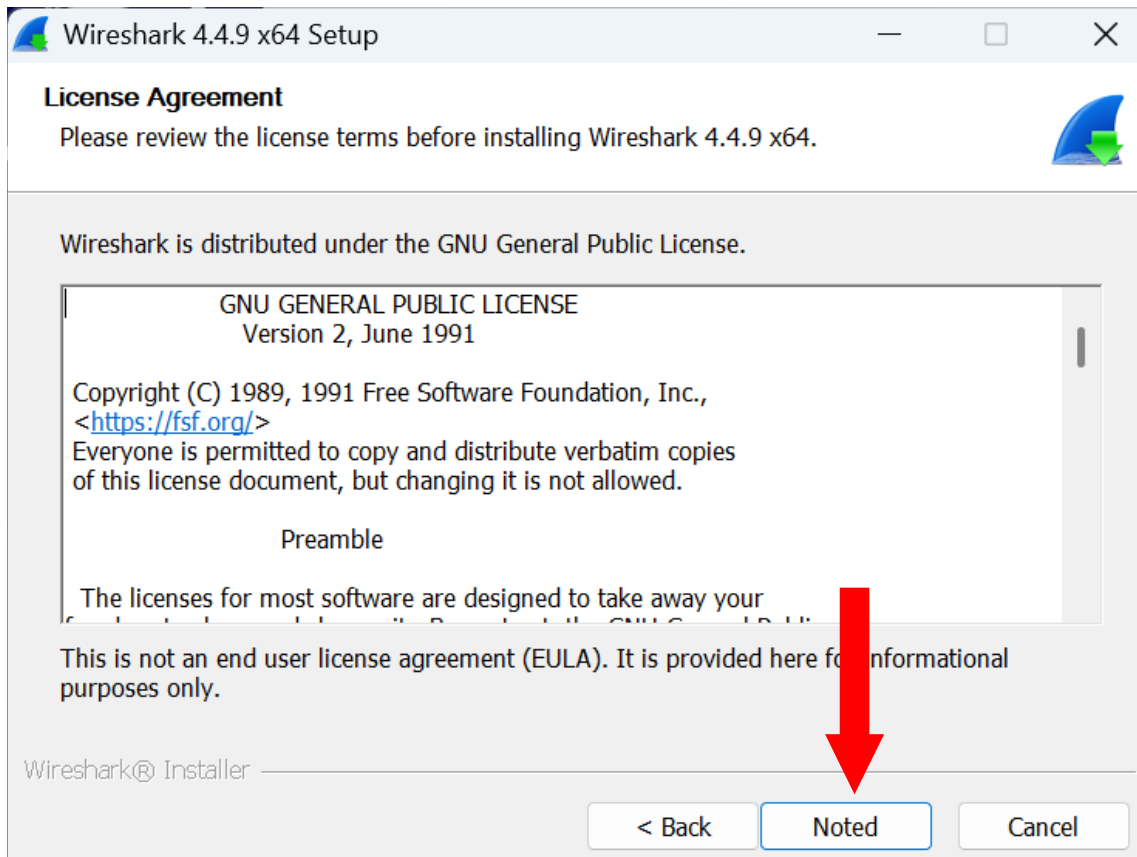
Go to the Wireshark download website: <https://www.wireshark.org/download.html>



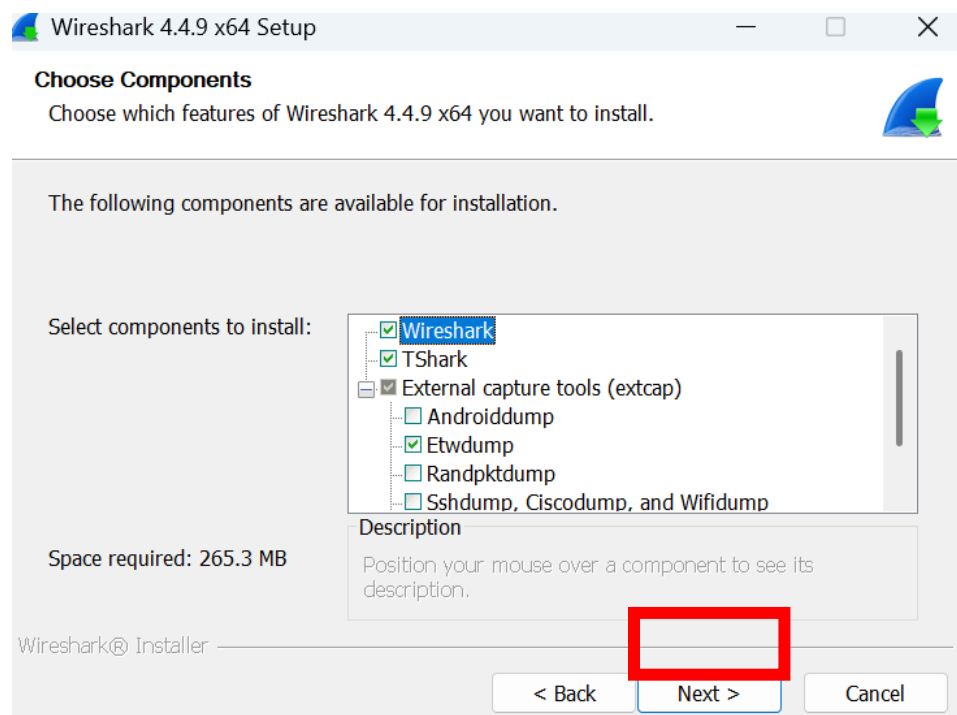
Select Next on the set-up window:



Select "Noted" once on the License agreement page

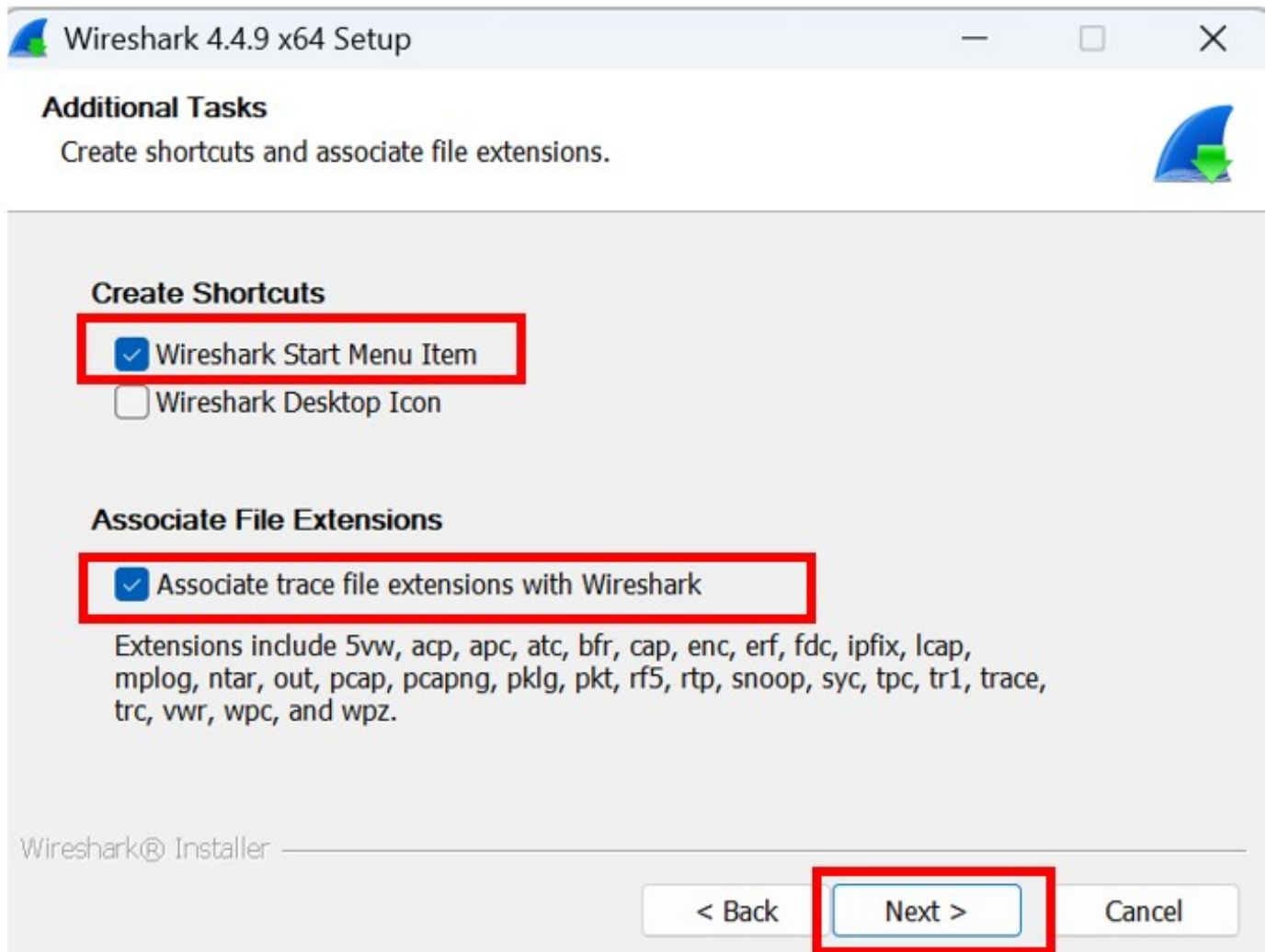


On the next screen select "Next":

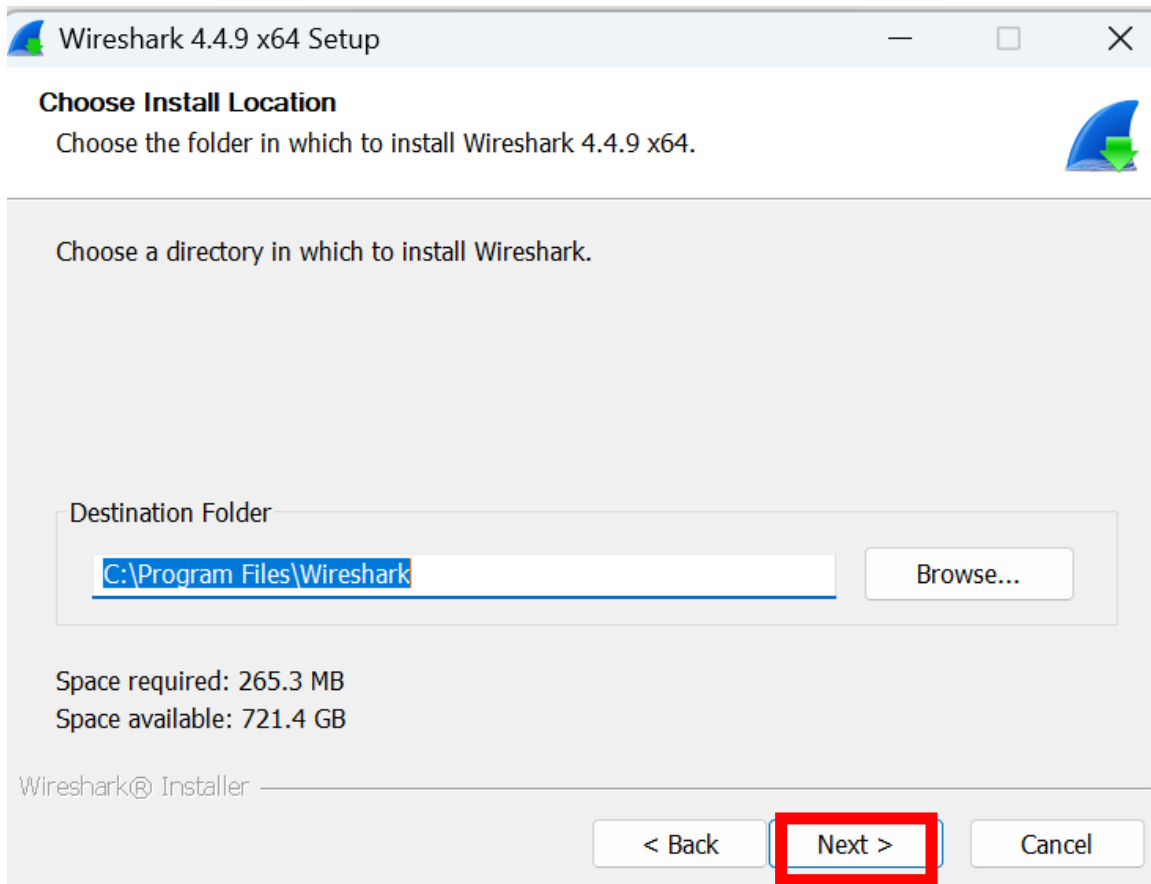


Select “Wireshark Start Menu item”, in the Create shortcuts section.

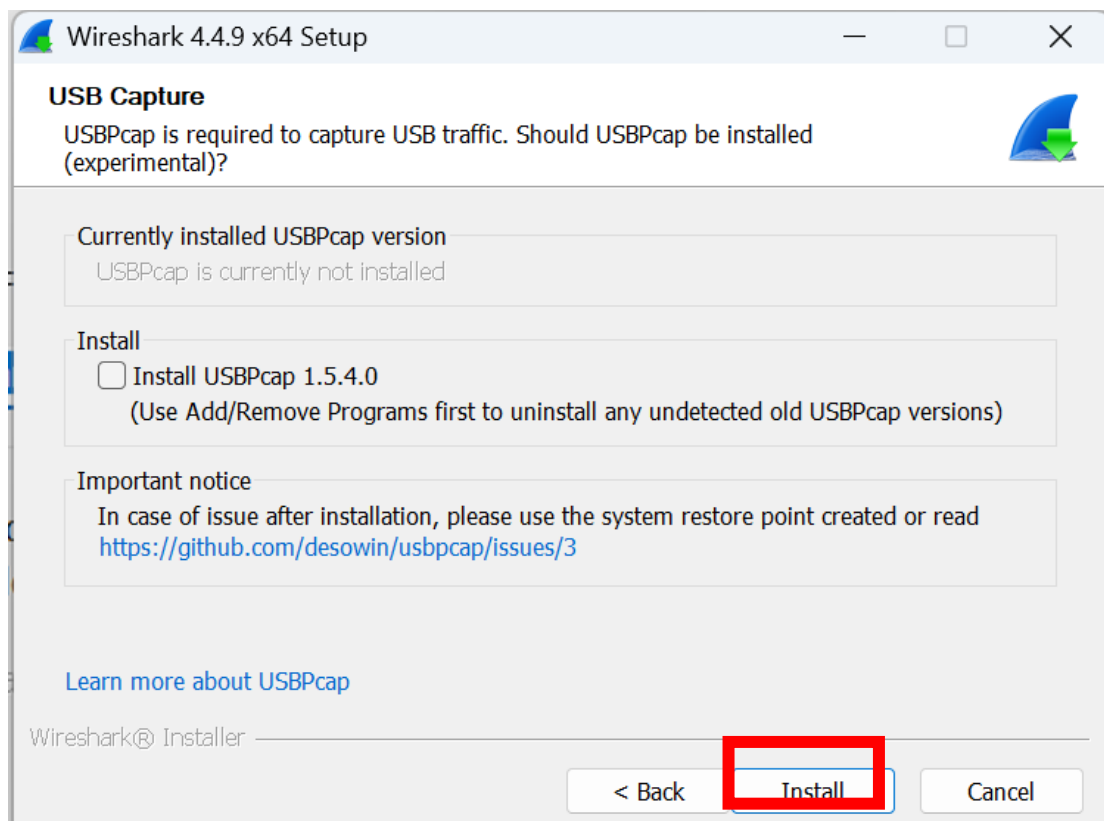
Select Associate trace file extensions with Wireshark in the Associate File Extensions section.



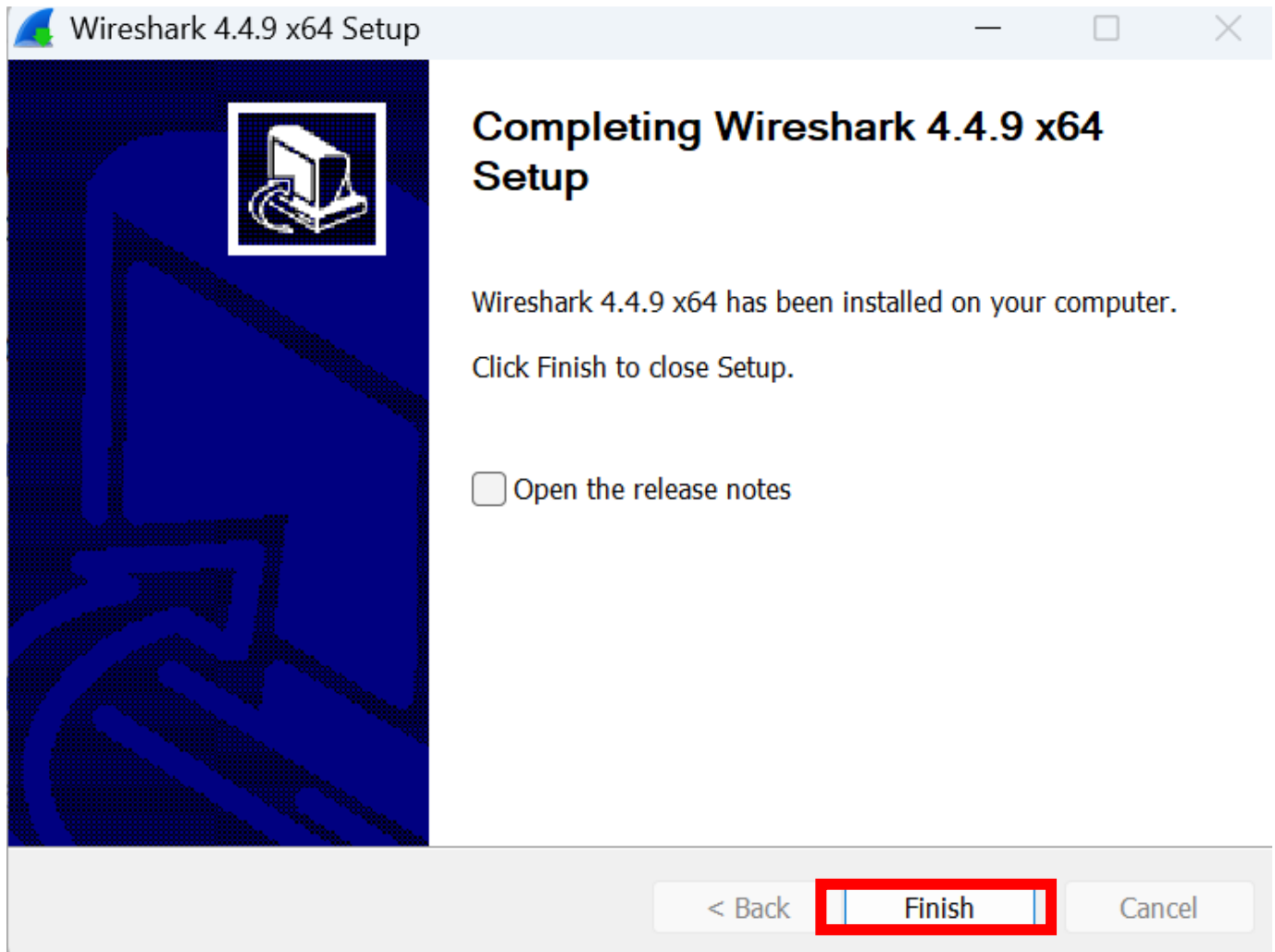
Select local location to install, select “Next”



After clicking next, select “Install”



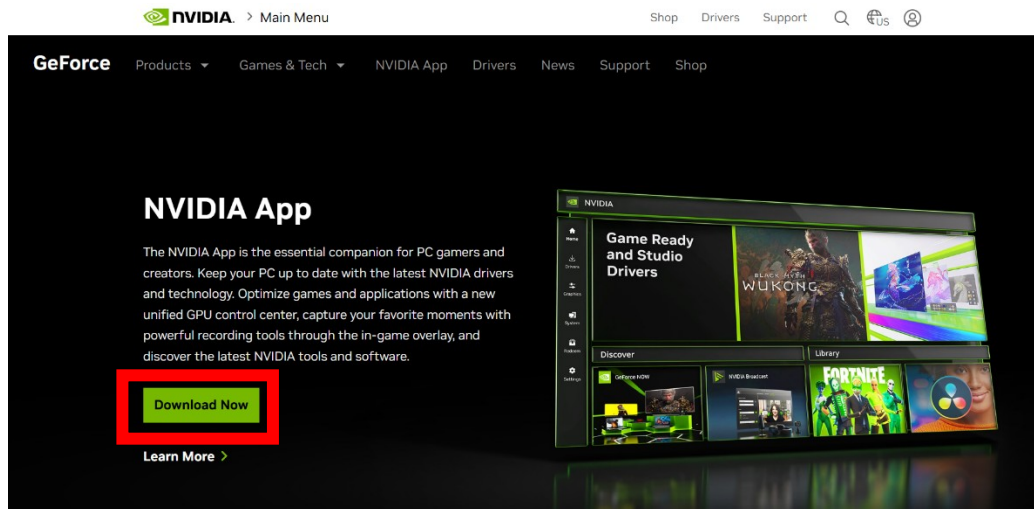
After selecting next you have completed Wireshark setup, select “Finish”



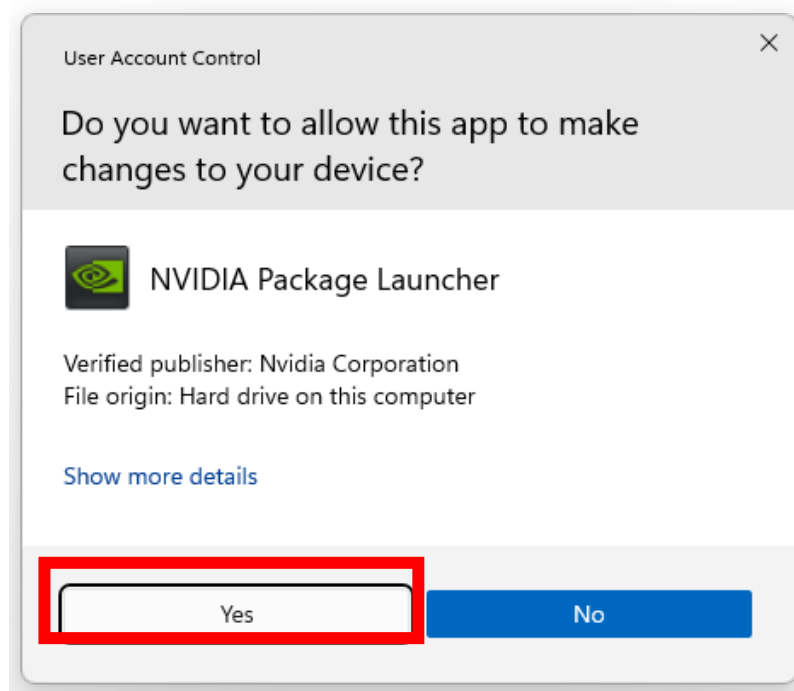
Installing NVIDIA Drivers:

Go to the download page -> <https://www.nvidia.com/en-us/software/nvidia-app/>

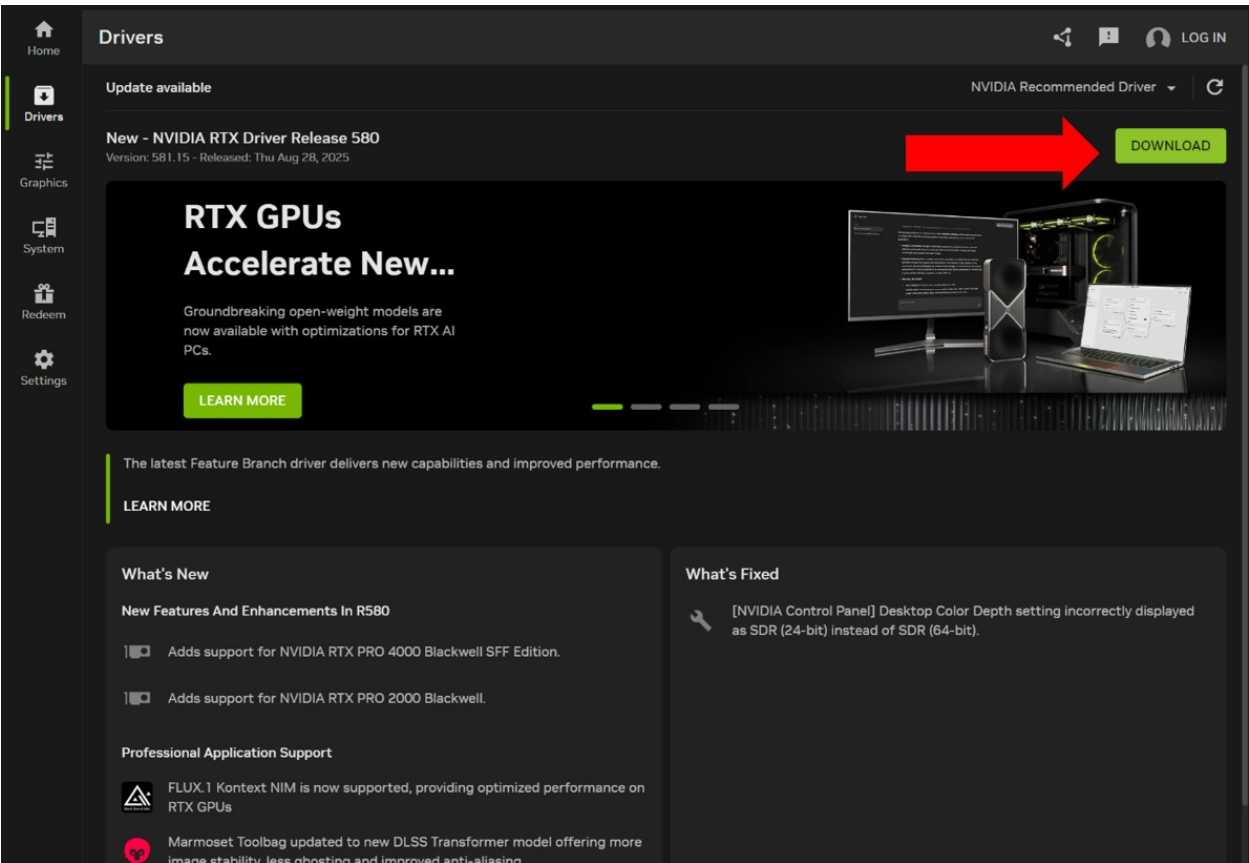
Select "Download Now"



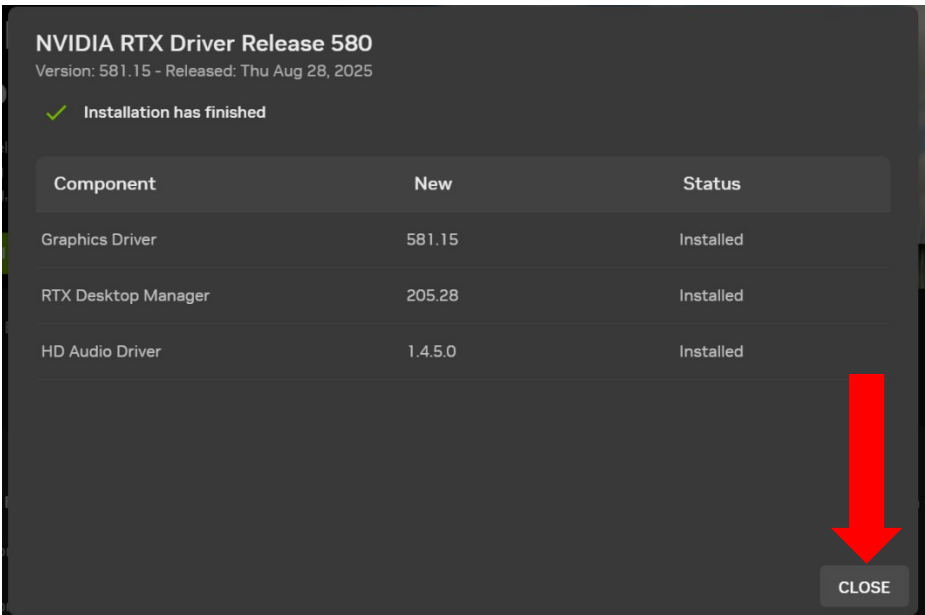
Allow Nvidia to make changes to your device:



Go to the Drivers tab, then click on download for the recommended driver



After a few minutes your driver will finish installing, after clicking close



Problem:

Problems that could occur:

Installer Freezes or Gets Stuck, Common during "Getting files ready for installation" or first boot.

Possible Causes: Bad USB drive or ISO, Faulty RAM or SSD

Conclusion:

In conclusion, this lab was important since it provided hands-on experience with latest operating system and tools. Being able to create clean, standardized system image on new devices is also crucial since it saves time during deployment in real scenarios. Overall, setting up a Windows 11 with essential applications ensures a practical, controlled environment for learning, testing, and preparing systems for real-world use.